Cross-Domain Coupling Network With Lightweight Fully Featured Mapping and Loop Aggregation for Semantic Segmentation of High-Resolution Remote Sensing Images

Xiaofeng Wang, Bangwei Chen, Yan Chen, Qianchuan Zhang, Kehong Wang, Lixiang Xu, Chen Zhang, and Le Zou

***Abstract*—To fully leverage contextual information for the precise segmentation of objects in remote sensing images, while addressing the challenges associated with substantial object scale variations and complex backgrounds, we propose a lightweight cross-domain coupling network (LCCN) tailored for the semantic segmentation of high-resolution remote sensing images (HRSIs). To standardize feature selection and fusion procedures, the LCCN incorporates an innovative encoder–coupler–decoder architecture designed to facilitate key feature extraction and optimization. A cross-domain coupling module (CDCM) is created in the coupler to conduct preliminary feature screening of spaces and dimensions based on channel and spatial attention. It performs multiscale feature extraction and global information modeling through feature grouping and loop aggregation. This helps to extract key features while reducing the computational overhead. To further decrease the interference from complex backgrounds, a secondary optimization of the key features is carried out: a lightweight fully featured mapping attention module (LFMAM) is designed within the decoder. LFMAM utilizes an interactive fusion strategy and a lightweight linear self-attention (SA) mechanism, comprehensively considering all interactions between global-to-global, global-to-local, local-to-local, and local-to-global processes. By capturing the effective correlations and variances among features to further refine them, it enables the network to further optimize the crucial information while ensuring lightweight. We have conducted extensive comparison experiments and ablation experiments on the ISPRS Vaihingen and ISPRS Potsdam datasets. The extensive experimental results demonstrate that our proposed LCCN can obtain superior performance compared to other advanced semantic segmentation models.**

***Index Terms*—Feature extraction and optimization, lightweight encoder–coupler–decoder, linear self-attention (SA) mechanism, loop aggregation, optical remote sensing semantic segmentation.**

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I. Introduction

REMOTE sensing technologies, encompassing satellite, aerial, and unmanned aerial vehicle (UAV) remote sensing, use sensors to detect electromagnetic radiation emitted from the Earth's surface and transform it into visualizable images. Owing to the incessant advancement of satellite and remote sensing technologies, it has become facile to acquire a vast quantity of high-resolution remote sensing images (HRSIs). The semantic segmentation of HRSIs can furnish detailed classification information regarding ground features, such as buildings, roads, and vegetation. This provides highly accurate data for tasks such as urban planning [1], [2], land surveying [3], [4], [5], disaster assessment [6], [7], and environmental monitoring [8]. Consequently, it enables efficient land management and accurate urban planning while also furnishing robust support for environmental protection and surveillance.

With the rapid development of deep learning, convolutional neural networks (CNNs) have emerged as dominant in the field of semantic segmentation due to their powerful feature extraction capabilities. The fully CNN (FCN) [9] realizes an end-to-end semantic segmentation network, serving as the cornerstone of fully convolutional semantic segmentation approaches. However, the simplicity of the decoder used in FCN leads to an inability to use the semantic information contained in the image. Subsequently, researchers have proposed numerous methods to improve the semantic segmentation network of FCN. UNet [10] modifies the network structure based on FCN, utilizing skip connections to link the encoder and decoder, thus fully leveraging the hierarchical features extracted by the fully convolutional network. PSPNet [11] introduces the pyramid pooling module (PPM), which establishes long-range dependencies by pooling features of different sizes and stacks them together. The DeepLab series [12],

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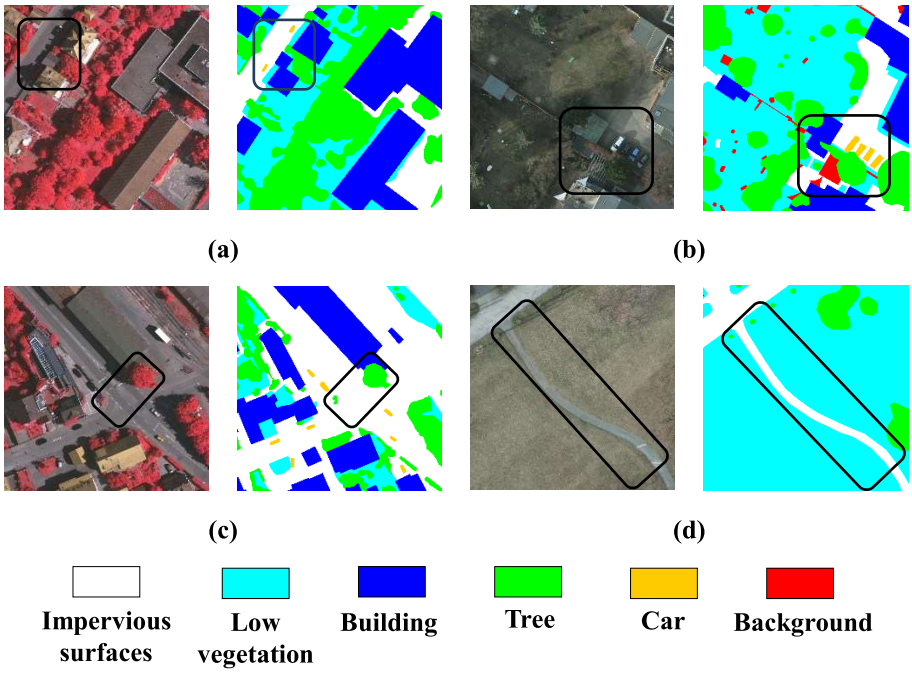


Fig. 1. Main challenges in semantic segmentation of HRSIs. (a) Shows that shadows cause significant differences within the same class of objects. (b) Shows a problem of large intra-class feature differences and small inter-class feature differences. (c) Shows that the size of the target objects in the remote sensing image varies. (d) Shows that some buildings and roads in the remote sensing image usually present irregular shapes.

[13], [14] networks enhance segmentation performance by using different dilation rates to gradually expand the receptive field of convolution, thereby better learning contextual information. Notwithstanding the notable performance that these methods display regarding overall segmentation accuracy, because of the convolution's dependence on a fixed receptive field, they are more adept at extracting the local feature information nestled within remote sensing images. This limitation restricts their capacity to capture a more extensive context and establish long-range feature interrelationships.

Compared to natural images [15] and medical images [16], remote sensing images exhibit more complex characteristics and substantial scale variations in the distribution of labeled objects within a limited scene. The main challenges of semantic segmentation in HRSIs are illustrated in Fig. 1, where the left-hand side of each image pair represents an HRSI and the right-hand side shows the corresponding pixel semantic categories denoted by different colors. For instance, in the area outlined by the rounded rectangle in Fig. 1(a), shadows cause significant variation within the same category of objects, potentially leading to misclassification of some targets in remote sensing images. Furthermore, as shown in Fig. 1(b), at the edges where different categories meet, the pixel features of each category lack distinctiveness. This consequently leads to issues characterized by significant intraclass feature variances and minimal interclass feature differences. In Fig. 1(c), owing to the distant capture, the sizes of some target objects exhibit variations. This may lead to the omission of small-scale objects during the segmentation procedure. In addition, certain buildings and roads in remote sensing images often present irregular shapes and may span the entire image, as illustrated in Fig. 1(d). Designing efficient architectures and fully extracting and utilizing contextual information and multiscale information are among the approaches to addressing these challenges.

Attention mechanisms based on CNNs augment the model's capacity to learn global features by concentrating on regions of interest within an image via global information processing. Attention mechanisms grounded in information aggregation initially aggregate features in accordance with the dimensions requiring attention. Subsequently, linear transformation operations and nonlinear activation functions are applied to the aggregated information to derive the attention weights. For example, Hu et al. [17] put forward the squeeze-and-excitation (SE) network, which strengthens the interrelationships between channels via SE blocks. The SE block is composed of two phases, squeezing and excitation, for the adaptive recalibration of features. It accentuates important features and suppresses less prominent ones by leveraging global information. ECANet [18] illustrates the substantial influence of channel dimensionality reduction on attention mechanism networks, presenting a local cross-channel interaction strategy that eschews dimensionality reduction. This strategy adaptively modifies the weights of channel features according to channel correlations, enabling the network to better focus on important features. The cross-channel interaction that extracts neighboring contextual information not only boosts network performance but also remarkably reduces the number of parameters and the computational burden. Another category of attention mechanism establishes similarity correlations between unit pairs via matrix multiplication to derive attention weights. For instance, Fu et al. [19] devised the DANet model integrating positional and channel attentions, drawing inspiration from the concept of nonlocal [20].

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It separately utilizes spatial pixels and channel features as query expressions for contextual modeling, enabling it to more effectively capture the interdependencies and contextual details among different pixel positions. Huang et al. [21] employ a novel crisscross attention module to aggregate the contextual information for each pixel in an interwoven manner. Consequently, it efficiently gains the contextual information that encompasses the whole image. Li et al. [22] propose a novel kernel attention mechanism that reduces computational demands and effectively extracts and utilizes global dependencies. Although these attention models have achieved significant progress, they often introduce a considerable amount of redundancy when extracting global contextual information. Moreover, when extracting spatial contextual information, they do not consider the interrelationships among different dimensions.

While the attention mechanisms based on CNNs have achieved breakthroughs in the field of remote sensing images' semantic segmentation, their ability to capture global contextual information remains limited. Notwithstanding the enhancements, such as the employment of attention mechanisms that partially mitigate this shortcoming, they do not fundamentally address the inherent limitations of CNNs. In marked contrast, the vision Transformer (ViT) [23] partitions the input image into fixed-sized image patches and processes these patches through self-attention (SA) mechanisms. As a result, it is able to capture the global interdependencies among the image patches. This global receptive field empowers ViT to model long-range dependencies with greater flexibility. Subsequently, a multitude of researchers have initiated the exploration of applying Transformer models to semantic segmentation tasks. In the realm of semantic segmentation, Transformer-based models can be broadly classified into two types. The first category is fully constructed upon the Transformer structure, such as Segmenter [24] and SwinUnet [25], which leverage the advantages of the Transformer to better consider global semantic features. They enhance contextual information interaction to extract features, thereby improving model segmentation accuracy. However, due to the quadratic complexity of the SA module in Transformer, they require high computational and time costs. The second category employs a hybrid structure of Transformer and CNNs, fully utilizing the powerful global contextual modeling capabilities of Transformer and the local detail extraction ability of CNNs, thus achieving better segmentation results. For instance, CCTNet [26] combines local details such as edges and textures captured by CNNs with the global contextual information obtained through a Transformer for remote sensing image crop segmentation. In addition, different modules such as test-time augmentation and postprocessing steps are introduced to remove holes and small objects during inference, restoring the complete segmentation image. Zhang et al. [27] propose a hybrid architecture that integrates the Swin Transformer [28] backbone, capable of capturing long-range dependencies, with a U-shaped decoder. The decoder uses a depthwise separable convolution-based accumulated spatial pyramid pooling block and SE modules to more effectively identify local details within the image. Gao et al. [29] propose a stagewise fusion model that can fully utilize coarse and fine feature information at different semantic scales, resulting in a model with good feature representation. An attention fusion module (AFM) is used to adaptively fuse the output features of the Transformer and CNN, providing rich semantic and spatial contextual information. In addition, there are methods analogous to Mamba, which are based on the state space model (SSM) [30], [31], [32]. For example, Zhou et al. [32] put forward MaDiNet, incorporating a specifically designed MambaSAR module. This module serves to capture rich and comprehensive spatial structural information of targets, highlight the positional attributes of targets within images, and consequently strengthen the detector's capability to discriminate between targets and complex background interferences. At present, semantic segmentation networks with standard CNN architectures cannot screen key features. Conventional attention mechanisms based on CNNs encounter significant challenges in modeling long-range dependencies. Semantic segmentation featuring pure Transformer architectures exhibits high computational complexity. In addition, architectures that are an encoder–decoder hybrid of CNNs and Transformers are insufficiently optimized in terms of the feature fusion phase. For instance, when a CNN serves as the encoder and a Transformer acts as the decoder, there is a shortage in the extraction of key features. Conversely, when a Transformer is used as the encoder and a CNN as the decoder, relatively high computational complexity will be introduced.

Therefore, with the aim of enhancing the extraction of key features of objects with varying scales in HRSIs, mitigating the interference from intricate backgrounds, optimizing the feature fusion process, and concurrently reducing the complexity of the entire model, we propose a cross-domain coupling network with lightweight fully featured mapping and loop aggregation for the semantic segmentation of HRSIs. By leveraging contextual information to assist in discrimination and combining semantic and spatial information with multiscale features, this network aims to resolve the issues of small interclass differences and large intraclass differences within the remote sensing images. It also strengthens the ability to model global information, mitigating the impact of shadowed areas and complex backgrounds on the segmentation results, and enhancing the segmentation capabilities for target objects. This proposed network integrates the global and local components to capture crucial information from the whole image, enabling the differentiation of various classes of objects and enhancing the segmentation efficacy, such as accuracy and complexity. The main contributions of our work are summarized as follows.

1) We propose a lightweight fully featured mapping and loop aggregation cross-domain coupled network for HRSI semantic segmentation. Rather than employing the conventional encoder–decoder structure, this network features an innovative encoder–coupler–decoder architecture. The architecture standardizes the processes of the initial screening, fusion, and refined optimization of key features while further addressing the challenges of scale variations and complex backgrounds in remote sensing images.

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2) To more effectively extract key features, within the proposed coupler, we design a cross-domain coupling module (CDCM). This module optimizes the information aggregation process in our enhanced conventional attention mechanisms. It selectively concentrates on feature channels and spatial information across diverse dimensions. Through the utilization of feature grouping and loop aggregation, it extracts multiscale features and models global information for each subfeature map. This approach not only reduces computational costs but also enriches feature information and improves the accuracy of target interpretation.
3) To address the issues of redundancy in conventional attention mechanisms and the high computational complexity of SA, we devise a lightweight fully featured mapping attention module (LFMAM) within the decoder. LFMAM utilizes an interactive fusion strategy and a lightweight linear SA mechanism. It comprehensively takes into account the processes from global to global, global to local, local to local, and local to global. This promotes the synergistic effect between different branch features and further refines feature correlation and difference while significantly reducing the computational complexity, making it more lightweight.

The structure of this article is as follows. Section II introduces the related work on the semantic task in remote sensing images. Section III provides detailed descriptions of the network constructed in this article. Section IV describes the datasets and evaluation metrics used in the experiments. Section V presents the experimental results and discussion. Section VI provides the conclusion of this article.

II. Related Work

A. Semantic Segmentation of the Encoder–Decoder Architecture

The extraction and delineation of surface-level information pertaining to terrestrial objects from HRSIs constitute a fundamental task within the domain of intelligent image interpretation. This task necessitates not only the capture of broad geographic features (e.g., terrain and land use types) but also the accurate identification and analysis of intricate details (e.g., building contours, road networks, and vegetation distribution). In comparison to natural images, the semantic segmentation of remote sensing images presents a series of distinct challenges. A substantial similarity exists between different categories, and notable variations may occur within a single category. The presence of cast shadows diminishes the variance within shadowed areas, thereby amplifying the challenges associated with semantic segmentation of remote sensing images. Early encoder–decoder architectures based on CNNs have achieved success in semantic segmentation tasks [9], [10], [11]. However, the spatial constraints imposed by convolutional operations lead to limited receptive fields, making it difficult to capture large-scale contextual information, ignoring long-distance dependencies, and lacking attention to scale variations.

To address these issues, researchers proposed various improvement strategies, including the use of contextual information, dilated convolutions, large-kernel convolutions, and multiscale strategies, combined with attention mechanisms to enhance the ability of CNN-based models to effectively capture contextual information. For example, Li et al. [33] highlight that the extensive field of view (FOV) inherent in remote sensing images presents a significant mechanism Multi-Scale Attention Fusion Module (MSAF) to establish connections between different branches, enhancing information transfer between features at different levels. ResUNet-a [34] ingeniously integrates the merits of the UNet and ResNet architectures. It embeds dilated convolutions and spatial pyramid modules to effectively capture multiscale contextual information within remote sensing images. As a result, it significantly bolsters the performance of semantic segmentation tasks. Guo et al. [35] combined dilated convolutions with dense convolutions to extract detailed information, eliminate the grid effect, and address the drawback of maintaining a balance between context and detailed information. They also utilized a MSAF to establish connections between different branches, enhancing information transfer between features at different levels. Li et al. [36] strategically utilize asymmetric convolution blocks to substitute the standard convolution layers, aiming to augment the feature representation capabilities. In addition, they devised multiscale skip connections integrated with channel attention mechanisms to amalgamate and refine the semantic features engendered at diverse levels. Although these strategies significantly improved the performance of the networks, they still overly rely on convolution operations and do not directly encode global context. Transformer, with its powerful global modeling capabilities, has attracted widespread attention and is gradually surpassing CNNs in the field of image segmentation. Researchers are progressively exploring semantic segmentation approaches for remote sensing images based on Transformer architectures. DC-Swin [37] utilizes Swin Transformer [28] as an encoder and proposes a densely connected feature aggregation module (DCFAM) to extract semantic features enhanced with multiscale relationships. It employs shared spatial attention (SSA) and shared channel attention (SCA) to enhance the spatial and channel relationships of semantic features, with downsampling connections and upsampling connections further integrating multilevel features, thereby enriching multiscale information and context. SFCRNet [38] employs a context refinement module to optimize the information aggregation procedure. It achieves this by generating weights and implementing multiplication and summation operations. This approach reinforces information aggregation at diverse locations, boosting the representativeness of the aggregated information and maximizing the reflection of the correlations and distinctions among the original data. Zheng et al. [39] propose a semantic decoupling transformer (SDT) that divides pixels into different semantic clusters using K-means to obtain rich scale information semantic features. By calculating the similarity between semantic features and image features, the SDT enhances the intraclass compactness and interclass separability. Consequently,

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it excavates the interdependencies among intraclass features and diminishes the correlation among features of diverse categories, thus improving the result of semantic information scale coupling in remote sensing images. CLCFormer [40] puts forward a bilateral feature fusion module (BiFFM) for the aggregation of extracted semantic features across multiple levels. It utilizes an attention gate (ATG) module to enhance feature representation, thereby improving the segmentation results of various-shaped targets. Furthermore, it uses an attention residual (ATR) module to optimize the learning of spatial features, mitigating the blurring of boundaries of occluded objects.

In semantic segmentation of remote sensing images, many works use U-shaped encoder–decoder structures and skip connections to fuse the multiscale information obtained by the encoder in the decoding stage. However, relying solely on a single instance of fusion might not be entirely sufficient. On the one hand, the low-level feature maps are found to have a rather significant amount of redundant nontarget noise. On the other hand, there appears to be a certain degree of deficiency in the representative key features within the high-level feature maps. It is essential to conduct a secondary screening and fusion optimization of the low- and high-level features from the encoder along the path based on skip connections. Therefore, we design the new encoder–coupler–decoder architecture to better perform feature screening and feature fusion.

B. Lightweight Semantic Segmentation

To augment the segmentation performance of networks, larger encoders are frequently employed for feature extraction, and intricate decoders are devised to integrate features. This results in a substantial upsurge in the number of parameters and the computational burden of the segmentation network. To adapt to scenarios demanding high efficiency and having restricted computational capabilities, it is essential to develop lightweight yet effective segmentation models. Besides reducing network dimensions and computational complexity, lightweight segmentation models must also consider two aspects: capturing contextual information and retaining as much detailed information as feasible to guarantee segmentation performance.

To meet the needs of real-time applications, researchers proposed numerous lightweight segmentation networks. For instance, BiseNet [41] and BiseNet-V2 [42] adopt a dual-path architecture, with one path for contextual sampling to obtain rich semantic information and the other path utilizing simple convolution operations to extract local details. This design not only achieves the effect of capturing contextual and local details but also ensures the network is light enough. ABCNet [43] employs the same bilateral architecture as BiseNet and introduces linear attention into the lightweight segmentation model to enhance the feature output of the contextual path. In addition, it designs a feature aggregation module with linear attention as its core to fully aggregate the features output from the contextual and spatial paths. Wang et al. [44] propose an efficient global–local attention mechanism to model global and local information in the decoder. The cross-shaped window context interaction module in the global attention branch can model long-range dependencies at the window level, thus capturing global context. The extracted global and local information is processed by the feature refinement module to obtain useful results. Chen et al. [45] put forward an adaptive mixed attention network. In this network, the channel attention module makes use of multiscale strategies and channel squeezing techniques to extract adjacent channel information and category information. Meanwhile, the spatial attention module extracts and fuses detailed information from diverse spatial dimensions. Ultimately, the adaptive fusion module explores the complementary relationship between channels and space, as well as the correlation between pixels and categories. CMTFNet [46] proposes a lightweight W/P Transformer module as the decoder, using window-based MHSA (W-MHSA) and pooling-based MHSA (P-MHSA) instead of the traditional MHSA in the Transformer to obtain global contextual information at a lower computational cost and connect with the encoder through skip connections. It uses a channel and spatial AFM to adaptively fuse deep semantic features and shallow detail features. DDRNet [47] introduces a meticulously designed detail-aware attention (DAA) module. By leveraging the cross correlation matrix between high-level features and the original image, it quantifies the similarity among objects of the same category, thus conveying rich semantic information from high-level features to small objects and compensating for the loss of small targets in the feature propagation process. AMMUNet [48] proposes a granularity multihead SA (GMSA) module in the decoder to extract attention weights and uses an attention map merging mechanism (AMMM) to merge multiscale attention maps. The use of granularity alignment and relative position bias reduction strategies effectively reduces computational demands.

Even though the aforementioned methods sustain a relatively compact model size, there remains substantial room for enhancement in terms of accuracy. In addition, the current feature mapping generally involves a process from the global to the local or from the local to the global. More diverse feature mappings need to be taken into consideration in order to obtain more comprehensive feature information. Simultaneously, in order to tackle the problems of feature redundancy inherent in conventional attention mechanisms and the elevated computational complexity associated with SA mechanisms, we propose and construct the LFMAM. Through the implementation of an interactive fusion strategy and a lightweight linear SA mechanism, we take into comprehensive consideration all the processes encompassing global-to-global, global-to-local, local-to-local, and local-to-global interactions. This approach fosters the synergistic interaction among features of diverse branches. While efficiently capturing the correlations and disparities among features and further refining the features, it drastically reduces the computational complexity of the model, rendering the model more lightweight and accurate.

III. Methodology

A. Overview of LCCN

The proposed semantic segmentation network, lightweight cross-domain coupling network (LCCN), is shown in Fig. 2, which adopts an innovative encoder–coupler–decoder architecture. We choose ResNet-18 as the encoder of the network,

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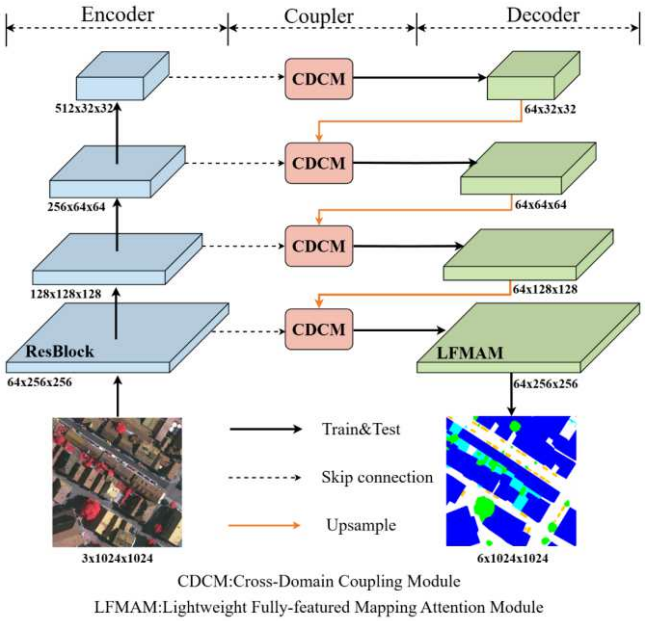


Fig. 2. Architecture of our proposed network (LCCN) for HRSI semantic segmentation. The network adopts an encoder–coupler–decoder framework, where ResNet-18 serves as the encoder, the CDCM acts as the coupler, and the decoder is integrated with the LFMAM. Input images are 1024 × 1024 (pixels) patches cropped from original high-resolution data.

and the coupler and decoder are composed of the CDCM and LFMAM proposed by us, respectively.

Due to the introduction of the residual connection mechanism in the ResNet [49], it not only effectively captures the rich multiscale semantic features in the image but also mitigates the problem of gradient vanishing and model degradation during network training. However, as the number of network layers continues to increase, a greater amount of the image's spatial detail information is lost. As a result, using a more-layered ResNet network as the main encoder does not yield a significant improvement compared to ResNet-18, especially in semantic segmentation tasks that are sensitive to spatial pixels. Therefore, when comprehensively considering the model's lightweight characteristics and efficiency, we choose the lightweight version ResNet-18, which is composed of four stages of Resblocks as the encoder of the network. Each stage conducts a downsampling operation on the feature maps with a scaling factor of 2. Meanwhile, the feature information of each stage is sequentially transmitted to the coupler through skip connections.

To enable more effective extraction and fusion of key features, we introduce a new module CDCM utilized to integrate the high-level semantic features with the low-level detailed spatial features. This module serves as a coupler and is composed of improved conventional spatial and channel attention mechanisms based on CNNs. It helps extract richer key features, and at the same time, it further ensures the lightweight nature of the model. This contrasts with the SA mechanism of the Transformer architecture. Through multiscale feature extraction and cross-dimensional spatial interaction coupling, the spatial details of the segmentation

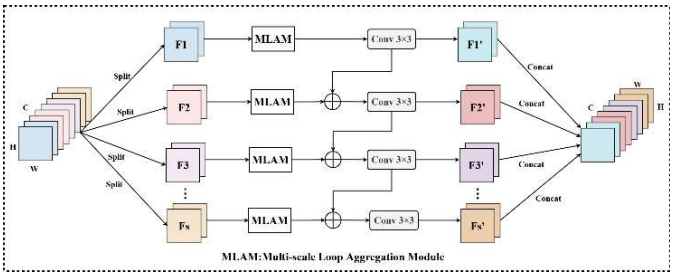


Fig. 3. Structure of CDCM.

outcomes can be enhanced. Subsequently, the fused feature map is fed into the decoder, which, in this study, is composed of the LFMAM designed and proposed by us. The LFMAM adopts an interactive fusion strategy and a lightweight linear SA mechanism. It comprehensively considers all processes from global to global, global to local, local to local, and local to global, promoting the synergistic effect among the features of different branches. While effectively capturing the feature correlations and differences and further optimizing the key features, it reduces the impact of complex backgrounds on the segmentation results, significantly decreases the computational complexity of the model, and makes the model more lightweight.

B. Cross-Domain Coupling Module

Owing to the intricate and diverse land features presented in remote sensing images, along with the significant variations in the sizes of the objects within them, it is infeasible to attain precise semantic segmentation of remote sensing images merely by relying on local context information. Currently, the majority of networks fail to fully exploit the effective feature information extracted at various levels of the main network. This leads to a deficiency of semantic information within the channels and the loss of spatial detail information, thereby exerting an impact on the segmentation performance of the networks. The CDCM uses feature grouping and loop aggregation to improve the ability of multiscale information extraction, attempting to obtain multiscale features at a finer granularity. Fig. 3 describes the proposed CDCM.

In Fig. 3, we can intuitively see that it is realized by four operations: split, multiscale loop aggregation module (MLAM), Conv, and Concat. First, the feature map $F \in R^{C\times H\times W}$ from the encoder and the upsampling fusion are used as the input for the CDCM, where C, H, and W represent the number of channels, height, and width, respectively. The split operation divides the image evenly into multiple subfeature maps along the channel dimension, with the set of subfeature maps represented as $Fi \in R^{c\times H\times W}(i \in \{1, 2, \ldots, s)$, each having the same spatial shape and the same number of channels, where $c = C/s$. Each subfeature map is then fed in parallel into the MLAM.

Fig. 4 shows the detailed structure of the MLAM, which uses three parallel paths to extract feature information from each subfeature map. Two of the parallel paths are designed to obtain spatial feature information of varying dimensions. This is achieved by applying average pooling to the input

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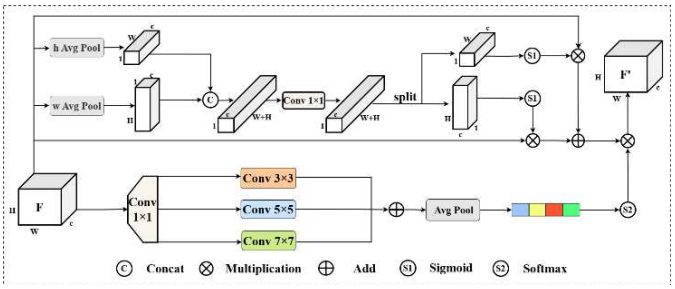


Fig. 4. Structure of MLAM.

feature map, thereby extracting the information of vertical and horizontal dimensions, resulting in feature maps $F_H \in R^{c\times 1\times W}$ and $F_W \in R^{c\times H\times 1}$, respectively. Then, F_H and F_W are concatenated along the height direction, and the features are fused via a 1×1 convolution operation. The resultant fused feature map is split into new feature maps F'_H and F'_W. Next, the sigmoid activation function is applied to generate attention weights, which are multiplied with the original subfeature maps through matrix multiplication. The updated two feature maps are added to achieve feature interaction across diverse dimensions between the two parallel paths. The third parallel path realizes multiscale aggregation through the summation of convolutions with different kernel sizes (3×3, 5×5, and 7×7). Subsequently, global average pooling is employed to reduce the dimension to $C \times 1 \times 1$, followed by a softmax function activation to generate channel attention maps, which are then multiplied with the spatially refined feature maps to obtain the updated subfeature maps. Thus, by means of global information encoding and cross-dimensional interaction, channel weights are recalibrated, and pairwise relationships at the pixel level are captured.

Each subfeature map refined by the MLAM has a set of corresponding convolution operations, consisting of standard 3×3 convolutions and batch normalization, represented as $A_i(\cdot)$, as shown in the following equation:

$$F'_i = \begin{cases} A_i(F_i), & i = 1 \\ A_i(F_i \oplus F_{i-1}), & 1 < i \le s \end{cases} \tag{1}$$

where $i \in \{1, 2, \ldots, s\}$, s represents the number of subfeature maps, F_i represents the subfeature map after the split operation, F'_i represents the enhanced multiscale subfeature map, and $\oplus$ represents the feature summation operation. Finally, the enhanced multiscale feature maps can be obtained through concatenation, as shown in the following equation:

$$F' = \text{Concat}\left(\left[F'_1, F_2 \cdots F_s\right]\right) \tag{2}$$

where the Concat $(\cdot)$ function denotes the concatenation operation along the channel dimension and F' represents the whole enhanced multiscale feature map. It should be noted that each convolution operation $A_i(\cdot)$ performed via the CDCM can extract the feature information of the previous part of the subfeature map $F_j(j \le i)$.

Moreover, following each 3×3 convolution operation, the output feature map has a larger receptive field. The output with a smaller receptive field is capable of capturing more details within the scene, which is crucial for identifying key components of objects. On the other hand, the output with a larger receptive field can focus on larger objects. By effectively processing input feature maps with a multiscale approach, it promotes the extraction of local and global feature information. In addition, the feature grouping we use reduces the computational cost of our method by controlling the number of channels. The greater the value of s, the more potent the multiscale feature representation ability becomes, and the more abundant the extracted feature information will be.

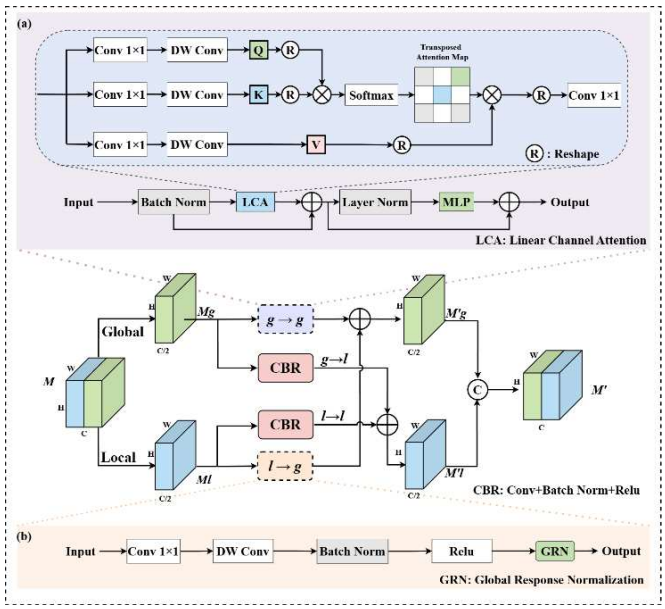


Fig. 5. Structure of LFMAM. Global features *Mg* and local features *Ml* are integrated through four different sub-branches respectively to achieve a global-local mixed receptive field. (a) represents the components of the $g \rightarrow g$ branch, and (b) represents the components of the $l \rightarrow g$ branch.

C. Lightweight Fully Featured Mapping Attention Module

To further address the issue of the interference caused by the complex backgrounds in remote sensing images to the actual performance of semantic segmentation networks and, meanwhile, reduce the background noise mixed in the key features extracted by the coupler, we propose the LFMAM. The LFMAM utilizes the interactive mixing of global–local receptive fields in a dual-branch structure. It simultaneously enables the interaction between global and local features, thus further addressing the issue that targets are prone to being interfered with by the background. As shown in Fig. 5, the LFMAM we proposed initially partitions the continuous feature map M into a global component Mg and a local component Ml along the feature channel dimension. Mg and Ml jointly are comprised of four distinct subbranches: global-to-global ($g \rightarrow g$), global-to-local ($g \rightarrow l$), local-to-global ($l \rightarrow g$), and local-to-local ($l \rightarrow l$). Each of these branches will be presented and explained individually as follows.

Currently, the mainstream approaches employed for acquiring global contextual information primarily consist of the following two methods. The first one involves incorporating an attention module at the tail end of the encoder, which may cause the network to lose more spatial detail information. The second method entails the direct utilization of the Transformer

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model within the network architecture. However, this not only increases the computational burden and the number of parameters but may also lead to the loss of local detailed feature information. Inspired by Restormer [50], we introduce a linear attention transformer module into the global branch of LFMAM to optimize the network's utilization of spatial information and capture global contextual information while effectively reducing computational costs.

The detailed structure of the $g \rightarrow g$ branch is illustrated in Fig. 5(a). The $g \rightarrow g$ branch is comprised of four operations: layer normalization, linear channel attention (LCA), element-wise addition, and a multilayer perceptron. The architectural design of this module bears a resemblance to that of the Transformer, which demonstrates remarkable proficiency in extracting global contextual information. When dealing with an input sequence, the SA mechanism in the conventional Transformer is required to calculate the relationships among all tokens present in that sequence. Each token must compute its attention weights to all other tokens, involving a softmax operation on an $N \times N$ matrix, where N is the sequence length. This computational approach leads to a quadratic increase in the computational complexity as a function of the sequence length. To achieve better global modeling capabilities while reducing high computational costs, we adopt an LCA to replace the multihead SA. The specific structure is shown in Fig. 5(b). For $Mg \in R \wedge (C/2 \times H \times W)$, we first achieve cross-channel pixel-level feature aggregation through a 1×1 pointwise convolution, followed by the use of a 3×3 depthwise separable convolution to extract channel contextual information. This reduces the number of parameters while retaining more spatial information. The query (Q), key (K), and value (V) are all generated through this process. This procedure can be formulated as follows:

$$Q = M^q_{\mathrm{DWconv}}\left(M^q_p\left(\mathrm{BN}\left(X\right)\right)\right) \tag{3}$$

$$K = M^k_{\mathrm{DWconv}}\left(M^k_p\left(\mathrm{BN}\left(X\right)\right)\right) \tag{4}$$

$$V = M^v_{\mathrm{DWconv}}\left(M^v_p\left(\mathrm{BN}\left(X\right)\right)\right) \tag{5}$$

where $M_{\mathrm{DWconv}}(\cdot)$ denotes the 3×3 depthwise separable convolution, $M_p(\cdot)$ denotes the 1×1 pointwise convolution, and BN denotes the batch normalization.

Subsequently, a reshape operation is performed on the query (Q) and key (K) to allow the dot product of query (Q) and key (K), followed by the softmax function to generate an attention map of size $R^{C \times C}$ that is multiplied by the value (V). Finally, a 1×1 pointwise convolution is used to generate the updated feature map. This procedure can be presented in the form of the following equation:

$$\mathrm{Attention}\left(Q, K, V\right) = V \cdot \mathrm{Softmax}\left(K \cdot \frac{Q}{\gamma}\right) \tag{6}$$

$$\hat{M} = M_p\left(\mathrm{Attention}\left(Q, K, V\right)\right) + X \tag{7}$$

where Softmax represents the function that generates the probability map, γ is a learnable scaling parameter that controls the dot product size of K and Q before the softmax function, and $M_p(\cdot)$ denotes a 1×1 pointwise convolution. M and $\hat{M}$ are the input and output feature maps. The above operation successfully reduces the complexity of attention computation to linear complexity, thereby effectively reducing computational costs.

For the $l \rightarrow g$ branch, as shown in Fig. 5(b), this branch consists of five operations: 1×1 convolution, depthwise separable convolution, ReLU activation function, batch normalization, and global response normalization (GRN). The first four operations are used to extract local contextual information, while the last operation is used to reduce redundancy in the transformed features. GRN has three steps, including feature aggregation, normalization, and calibration [51]. It normalizes features on each channel to enhance their representational capabilities. In addition, both the $g \rightarrow l$ and $l \rightarrow l$ branches capture local features through the same operation, which consists of a 3×3 convolution, batch normalization, and ReLU activation function. $g \rightarrow g$ and $l \rightarrow g$ are added to obtain the interactive global feature $M'g$, and $g \rightarrow l$ and $l \rightarrow l$ are added to obtain the interactive local feature $M'l$. Finally, $M'g$ and $M'l$ are concatenated to produce the enhanced feature M'. Overall, the LFMAM process is defined as follows:

$$M'g = g \rightarrow g\left(Mg\right) + l \rightarrow g\left(Ml\right) \tag{8}$$

$$M'l = l \rightarrow l\left(Ml\right) + g \rightarrow l\left(Mg\right) \tag{9}$$

$$M' = \mathrm{Concat}\left(M'g + M'l\right) \tag{10}$$

where Mg and Ml represent the global component and the local component, respectively. The Concat$(\cdot)$ function denotes the concatenation operation along the channel dimension.

IV. Datasets and Experimental Setup

A. Datasets

In this study, we comprehensively evaluate our method using two remote sensing image semantic segmentation datasets, Vaihingen and Potsdam, both provided by the International Society for Photogrammetry and Remote Sensing. The Vaihingen dataset, captured in the town of Vaihingen, Germany, is renowned for its unique urban landscapes, including dense building complexes, complex road networks, and diverse vegetation cover. The Potsdam dataset covers a wide area of the city of Potsdam, Germany. It is characterized by high-resolution aerial imagery and has attracted significant attention due to its rich scenes. Both datasets contain six categories: impervious surface, building, low vegetation, tree, car, and background.

1) ISPRS Vaihingen Dataset: The Vaihingen dataset comprises 33 high-resolution images, with an average size of 2494×2064 pixels and a ground sampling distance of 9 cm. Each image provides three multispectral bands: near-infrared, red, and green. The dataset includes six categories. We use images with IDs 2, 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 27, 29, 31, 33, 35, and 38 for testing, image ID 30 as the validation set, and the remaining images as the training set. The original images are cropped to a size of 1024×1024 pixels.

2) ISPRS Potsdam Dataset: The Potsdam dataset consists of 38 high-resolution images, each measuring 6000×6000 pixels with a ground sampling distance of 5 cm. Each image provides four multispectral bands: red, green, blue, and near-infrared. The dataset includes six categories. We use images with IDs 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15,

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TABLE I
ABLATION STUDY OF EACH COMPONENT OF THE LCCN. THE BEST AND SECOND BEST VALUES IN THE COLUMN ARE IN BOLD AND UNDERLINED

Dataset	method	OA(%)	mF1(%)	mIoU(%)
Vaihingen	Baseline	91.63	88.20	79.38
	Baseline+LFMAM	92.67	90.03	82.24
	Baseline+CDCM	<u>92.81</u>	<u>90.23</u>	<u>82.54</u>
	Baseline+LFMAM+CDCM	**93.49**	**91.47**	**84.59**
Potsdam	Baseline	88.04	89.91	81.95
	Baseline+LFMAM	89.85	91.33	84.30
	Baseline+CDCM	<u>90.20</u>	<u>91.52</u>	<u>84.56</u>
	Baseline+LFMAM+CDCM	**91.30**	**92.57**	**86.39**

5_13, 5_14, 5_15, 6_13, 6_14, 6_15, and 7_13 for testing, image ID 2_10 as the validation set, and the remaining 22 images (excluding the misannotated image 7_10) as the training set. In our experiments, we use only the red, green, and blue multispectral bands and crop each original image to a size of 1024 × 1024 pixels.

B. Experimental Setup

We compared a diverse set of advanced network models on public datasets to verify the effectiveness of the proposed method, including convolution-, attention mechanism-, and Transformer-based methods. All experiments were performed on NVIDIA GTX 3090 GPUs with 24 GB of memory, using the PyTorch deep learning framework on an Ubuntu system. The AdamW optimizer was employed with a base learning rate of $6e^{-4}$, a weight decay of 0.01, and a cosine learning rate scheduling strategy. The batch size was set to 8, and all models were trained for 105 epochs (it should be noted that we set 15 epochs for the automatic update of the learning rate, so as to better align with the cosine learning rate scheduling strategy). The cross-entropy loss function is used.

For both the Vaihingen and Potsdam datasets, given the large size of the original dataset images, we cropped them into nonoverlapping 1024 × 1024 patches for training efficiency. We applied identical data augmentation techniques during training, including random scaling with factors [0.75, 1.0, 1.25, 1.5], random vertical and horizontal flipping, and random rotation.

C. Evaluation Metrics

The evaluation metrics used in our experiments fall into two main categories. The first is model efficiency, including the number of floating-point operations (FLOPs) for assessing complexity, frames per second (FPS) to evaluate the inference speed of the model, and the number of model parameters. The second category is the accuracy of the network, which includes overall accuracy (OA), mean $F1$-score (m$F1$), and mean Intersection over Union (mIoU). The calculation formulas for the evaluation metrics are presented as follows:

$$mF1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \times \frac{1}{N} \tag{11}$$

$$\text{mIoU} = \frac{1}{N} \sum_{k=1}^{N} \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k + \text{FN}_k} \tag{12}$$

$$\text{OA} = \frac{\sum_{k=1}^{N} \text{TP}_k}{\sum_{k=1}^{N} (\text{TP}_k + \text{FP}_k + \text{TN}_k + \text{FN}_k)} \tag{13}$$

where precision and recall are defined as follows:

$$\text{Precision} = \frac{1}{N} \times \sum_{k=1}^{N} \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k} \tag{14}$$

$$\text{Recall} = \frac{1}{N} \times \sum_{k=1}^{N} \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k} \tag{15}$$

where TP_k, TN_k, FP_k, and FN_k represent the true positive, false positive, and false negative, respectively, for objects indexed as class k. OA is calculated for all categories, including the background.

V. RESULTS AND DISCUSSION

A. Ablation Study

1) Components of LCCN: To verify the performance of each component proposed in this article, we conduct a series of ablation experiments on the IRPRS Vaihingen and IRPRS Potsdam datasets. To ensure fairness in the experiments, all parameters except the network structure are kept consistent in this round. Our baseline model is a simple U-shaped network with ResNet-18 as the backbone. Under limited memory conditions, the distribution of parameters and computational load between the backbone network and the encoder is of great significance. When they are relatively balanced, the overall performance of the network is expected to be better. Moreover, considering the lightweight characteristics, we ultimately chose ResNet-18 as the backbone network. The specific experimental results are shown in Table I.

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TABLE II
ABLATION STUDY OF FEATURE GROUPING AND MULTISCALE CONVOLUTION ON THE VAIHINGEN DATASET. THE BEST AND SECOND BEST VALUES IN THE COLUMN ARE IN BOLD AND UNDERLINED

Feature Grouping	Multi-scale Convolution	mIoU(%)	Parameters(M)	FLOPs(G)
×	×	79.38	11.38	**44.26**
×	√	80.20	12.42	51.58
√	×	<u>81.82</u>	**11.38**	<u>44.28</u>
√	√	**82.54**	<u>11.46</u>	44.73

We integrate LFMAMs into the baseline. By fusing local details with global contextual features, the network can capture more comprehensive semantic information, enhancing its ability to extract global contextual content. As shown in Table I, the addition of the LFMAM to the baseline network significantly improved the mIoU by 2.86% and 2.35% on the Vaihingen and Potsdam datasets, respectively.

After incorporating CDCM into the baseline network, the performance improvements on the Vaihingen and Potsdam datasets amount to 3.16% and 2.61%, respectively. CDCM aggregates multiscale attention for input feature maps hierarchically. Each subfeature map aggregates information along both height and width dimensions, followed by convolutional fusion to enhance the network's ability to extract multiscale information.

Both LFMAM and CDCM are merged into the baseline network, forming the proposed LCCN. LCCN outperforms all previous ablation modules, with the experimental results not only verifying the potency of LFMAM but also the significant role of CDCM. Ultimately, our method achieves the best results on the three metrics on both public datasets.

2) Effect of Feature Grouping and Multiscale Convolution: There are many targets with significant scale variations in remote sensing images. Convolutions with different sizes can capture richer multiscale information, and compared with single-scale convolutions, they can improve segmentation accuracy. We selected ResNet-18 as the baseline network and verified the importance of feature grouping and multiscale convolutions through ablation experiments. As shown in Table II, after adding the multiscale convolution module, the segmentation accuracy of the model increased by 0.82%. However, multiscale convolutions with different kernel sizes lead to an increase in computational complexity. Subsequently, the adoption of the feature grouping strategy enables the model to be distributed across multiple channels in more batch dimensions for processing. This not only effectively reduces the computational complexity but also strengthens the representation ability of target semantics through the mutual fusion of multiple groups of features. The experimental results show that compared with the baseline model, the segmentation accuracy is improved by 3.16%, and while maintaining lightweight characteristics, the model effectively utilizes multiscale features to enhance its adaptability to remote sensing images.

3) Effect of the Number of Feature Grouping: For a given input feature map, the CDCM partitions it into s subfeatures along the channel dimension to learn discriminative semantic information. To validate the correlation between s and model performance, we conducted ablation experiments with $s \in 2, 4, 8, 16, 32, 64$, systematically investigating the impact of the number of feature groups on model performance. The performance on the Vaihingen dataset is presented in Table III. Increasing the number of feature groups has a negligible effect on the total number of parameters but significantly improves the mIoU and OA. The model achieves peak performance when $s = 32$, while further increasing s leads to performance degradation. In-depth analysis reveals that feature grouping effectively reduces computational complexity through channelwise dimensionality reduction (from C to C/s per group), while parallel regional correlation calculations across multiple groups enhance the representation capacity of target semantics. However, excessive grouping ($s > 32$) results in insufficient channels per group, forcing the model to learn noisy features or converge to local optima, thereby introducing prediction bias. Based on these findings, we selected the optimal number of feature groups $s = 32$ in our experiments, striking a balance between feature expressiveness and computational efficiency.

4) Effect of MLAM: To validate the critical feature screening efficacy of the MLAM, we replace it with several other state-of-the-art attention mechanisms while keeping the other modules unchanged and conducted ablation experiments. These attention mechanisms include scSE [52], ECA [18], CBAM [53], SK [54], CA [55], and EGLA [44]. CDCM optimizes the information aggregation process in traditional attention mechanisms. It selectively focuses on feature channels and spatial information across different dimensions. By utilizing feature grouping and loop aggregation, it extracts multiscale features and constructs a global information model for each subfeature map. This approach not only reduces computational costs but also enriches the feature information and improves the accuracy of target interpretation. Due to the enhanced LCCN understanding of the image, LCCN consistently achieves the best OA, m$F1$, and mIoU on the Vaihingen dataset, as shown in Table IV.

5) Effect of LFMAM: Table V presents the ablation validation of each design component within LFMAM, comparing the experimental results of "global," "local," and their interaction mode "interaction" in LFMAM. Meanwhile, it includes a comparative analysis between the lightweight LCA adopted in this module and the traditional SA. We selected ResNet-18 as the baseline network. Table V further quantifies the impact of key components on the LFMAM module, with detailed analysis as follows. The model containing only the $g \rightarrow g$ (combined with LCA) achieves an mIoU of 81.81%, which is 2.1% higher than that of the model containing only the local branch (79.71%). This indicates that with the assistance of the LCA, $g \rightarrow g$ has a stronger ability to capture contextual information, while relying solely on $l \rightarrow l$ makes it difficult to achieve the same performance due to the lack of global

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TABLE III

ABLATION STUDY OF THE NUMBER OF FEATURE GROUPS ON THE VAIHINGEN DATASET. THE BEST AND SECOND BEST VALUES IN THE COLUMN ARE IN BOLD AND UNDERLINED

s	OA(%)	F1(%)	mIoU(%)	Parameters(M)	FLOPs(G)
2	93.09	90.58	83.14	11.88	62.45
4	93.16	90.69	83.33	11.69	55.05
8	93.29	90.84	83.57	11.63	51.35
16	<u>93.47</u>	<u>91.35</u>	<u>84.38</u>	11.59	49.50
32	**93.49**	**91.47**	**84.59**	<u>11.57</u>	<u>47.96</u>
64	**93.49**	91.25	84.25	**11.54**	**46.81**

TABLE IV

ABLATION STUDY OF DIVERSE ATTENTION MECHANISMS ON THE ISPRS VAIHINGEN DATASET. THE BEST AND SECOND BEST VALUES IN THE COLUMN ARE IN BOLD AND UNDERLINED

Attention Mechanism	IoU(%)					OA(%)	F1(%)	mIoU(%)
	ImSurf	Building	LowVeg	Tree	Car			
+scSE [52]	92.56	89.94	71.28	81.26	75.51	92.53	89.95	82.11
+ECA [18]	93.65	90.61	72.72	82.02	75.26	93.08	90.4	82.85
+CBAM [53]	93.52	90.75	<u>72.75</u>	82	74.91	93.05	90.36	82.79
+SK [54]	<u>93.82</u>	90.88	72.57	81.89	<u>76.62</u>	<u>93.15</u>	<u>90.59</u>	<u>83.16</u>
+CA [55]	93.58	<u>91.13</u>	72.4	<u>82.02</u>	76.58	93.07	90.58	83.14
+EGLA [44]	93.28	90.51	72.63	**82.42**	75.99	92.97	90.48	82.96
Ours	**94.23**	**92.01**	**74.19**	81.88	**80.64**	**93.49**	**91.47**	**84.59**

TABLE V

ABLATION STUDY OF EACH COMPONENT OF LFMAM ON THE VAIHINGEN DATASET. THE BEST AND SECOND BEST VALUES IN THE COLUMN ARE IN BOLD AND UNDERLINED

LFMAM			LCA	SA	OA(%)	mF1(%)	mIoU(%)	Parameters(M)	FLOPs(G)
global	local	interaction							
√	×	×	√	×	92.71	89.73	81.81	<u>11.48</u>	<u>47.19</u>
×	√	×	×	×	92.18	88.34	79.71	**11.41**	**46.91**
√	√	×	√	×	<u>92.84</u>	89.86	82.01	11.49	47.34
√	√	√	√	×	92.67	<u>90.03</u>	<u>82.24</u>	11.50	47.52
√	√	√	×	√	**93.00**	**90.32**	**82.72**	12.14	48.39

semantic guidance. Meanwhile, their parameter and FLOPs are nearly the same.

Second, there is a synergistic effect when $g \rightarrow g$ is combined with $l \rightarrow l$. When both $g \rightarrow g$ and $l \rightarrow l$ are added without interaction, the model's mIoU reaches 82.01%, which is 0.2% higher than that when only the $g \rightarrow g$ is used. This confirms the complementarity between global contextual information and local fine-grained features. Notably, after introducing global–local interaction, the mIoU further increases to 82.24% and rises to 90.03%. This verifies that the interaction mechanism enhances the feature refinement effect by enabling cross-feedback between global and local features, thereby improving segmentation accuracy.

Finally, there is a tradeoff between the LCA and the traditional SA mechanism. After replacing the LCA with the SA, the mIoU only increases by 0.48%, but the number of parameters increases significantly (from 11.50M to 12.14M, an increase of 0.64M), and the FLOPs also rise remarkably (from 47.52G to 48.39G, an increase of 0.87G). This result further proves that the LCA greatly reduces computational overhead with minimal precision loss, making it more suitable for HRSIs. In contrast, the performance improvement brought by the SA is negligible, but it is accompanied by excessive computational costs, which limit its practicality in large-scale input scenarios. The experimental results show that integrating global–local branches with feature interaction in LFMAM is crucial for improving segmentation performance, and the LCA

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TABLE VI
COMPARISON OF THE COMPLEXITIES OF OUR METHOD WITH OTHER METHODS. THE BEST AND SECOND BEST VALUES IN THE COLUMN ARE IN BOLD AND UNDERLINED

Method	Parameters(M)	FLOPs(G)	FPS	OA(%)	mF1(%)	mIoU(%)
PSPNet [11]	27.57	105.74	31	90.91	86.35	76.73
MAResU-Net [56]	26.28	105.10	89	<u>93.07</u>	90.58	82.84
MANet [22]	11.98	88.76	75	92.72	90.41	82.80
A2-FPN [59]	12.26	48.64	97	92.72	89.97	82.16
BANet [57]	12.73	50.91	25	92.86	90.43	82.89
ABCNet [43]	13.39	62.48	<u>113</u>	92.72	90.16	82.44
DC-swin [37]	66.90	267.79	16	92.88	90.60	83.18
UNetFormer [44]	<u>11.68</u>	**46.90**	**117**	92.82	90.37	82.78
CMTFNet [46]	30.68	134.48	70	92.88	90.57	83.16
AANet [45]	12.72	63.48	71	92.74	90.49	82.99
MSGCNet [58]	27.02	116.86	62	93.03	<u>90.82</u>	<u>83.51</u>
Ours	**11.57** (-0.11)	<u>47.96</u> (+1.06)	82 -	**93.49** (+0.42)	**91.47** (+0.65)	**84.59** (+1.08)

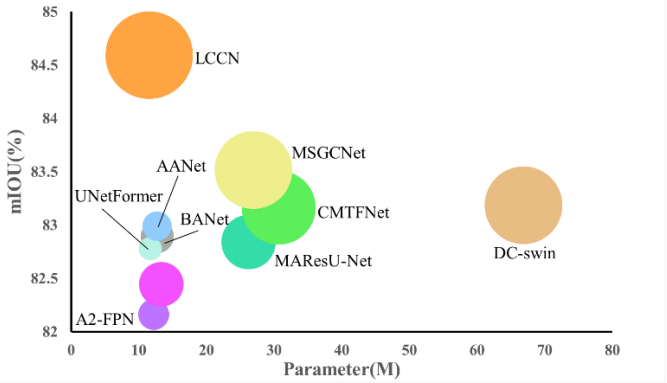


Fig. 6. Comparison of parameters (M), mIoU, and percentage of various models (results from the Vaihingen dataset).

achieves the optimal balance between efficiency and accuracy in remote sensing applications.

B. Comparison of Computing Complexity

In practical applications, the complexity and accuracy of the algorithm are important indicators for measuring its merits. The proposed network is compared with the currently most advanced lightweight networks in terms of the number of parameters, computational load, OA, m$F1$, mIoU, and FPS, with the comparison results shown in Table VI. Among these networks, UNetFormer has the smallest computational load, and compared with it, LCCN shows significant improvements in OA and mIoU. Specifically, compared with the UNetFormer network on the Vaihingen dataset, the mIoU, m$F1$, and OA scores of LCCN are 1.81%, 1.10%, and 0.67% higher, respectively. Moreover, we conduct a simple performance evaluation of LCCN, as shown in Fig. 6. Through a comprehensive analysis of Table VI and Fig. 6, LCCN strikes a good balance between performance and efficiency. Compared with other methods, it has a relatively low computational complexity and achieves the highest mIoU.

This finding can be explained by two reasons: first, CDCM achieves higher computational efficiency through feature grouping; second, LCCN introduces LFMAM to model long-range dependence relationships, using a global branch and local branch interaction fusion strategy to make the feature fusion more complete and improve the segmentation accuracy of the target. In general, compared with other models, LCCN attains the best balance between computational complexity, parameter size, and inference speed, which achieves the best performance, making it practical for real-world remote sensing applications where both accuracy and efficiency are critical.

C. Comparison With the State-of-the-Art Methods

To verify the superiority of LCCN, we compare it with some popular methods on the Vaihingen and Potsdam datasets. These approaches exhibit exceptional performance in multiscale feature extraction and global context modeling, substantially enhancing the semantic interpretation of remote sensing images. By integrating multiscale information and jointly learning diverse attribute features, these methods excel at capturing the rich and complex semantic signatures in remote sensing imagery, providing robust support for improving the accuracy and robustness of semantic interpretation.

The selected methods mainly include convolution-based networks (e.g., PSPNet [11], DeepLabV3+ [14], DSNet [35], and LoG-CAN [62]), attention mechanism-based networks (e.g., MAResU-Net [56], MANet [22], A2-FPN [59], ABCNet [43], and AANet [45]), and Transformer-based methods (e.g., BANet [57], DC-Swin [37], UNetFormer [44], CMTFNet [46], MSGCNet [58], WiCoNet [60], EMRT [61], and

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TABLE VII
QUANTITATIVE COMPARISON RESULTS ON THE IRPRS VAIHINGEN TEST SET WITH STATE-OF-THE-ART LIGHTWEIGHT NETWORKS. THE BEST AND SECOND BEST VALUES IN THE COLUMN ARE IN BOLD AND UNDERLINED

Method	IoU(%)					OA(%)	F1(%)	mIoU(%)
	ImSurf	Building	LowVeg	Tree	Car			
PSPNet [11]	90.86	85.49	69.56	78.64	59.10	90.91	86.35	76.73
DeepLabV3+ [14]	90.40	83.37	68.72	79.07	59.54	90.51	86.06	76.22
MAResU-Net [56]	93.58	91.13	72.40	<u>82.02</u>	76.58	93.07	90.58	82.84
MANet [22]	93.08	89.03	72.52	81.95	77.43	92.72	90.41	82.80
A2-FPN [59]	93.07	90.23	72.01	81.27	74.21	92.72	89.97	82.16
BANet [57]	93.50	90.54	72.03	81.22	77.18	92.86	90.43	82.89
ABCNet [43]	93.14	90.22	73.11	81.36	74.37	92.72	90.16	82.44
DC-swin [37]	93.40	91.05	71.48	81.53	78.42	92.88	90.60	83.18
UNetFormer [44]	93.56	90.23	72.55	80.72	76.85	92.82	90.37	82.78
CMTFNet [46]	<u>93.77</u>	<u>91.29</u>	70.67	80.93	79.13	92.88	90.57	83.16
DSNet [35]	93.57	91.03	<u>73.34</u>	81.66	77.54	<u>93.14</u>	<u>90.87</u>	<u>83.60</u>
AANet [45]	93.38	90.32	70.93	81.02	<u>79.29</u>	92.74	90.49	82.99
MSGCNet [58]	93.56	90.30	72.86	**82.50**	78.31	93.03	90.82	83.51
Ours	**94.23** (+0.46)	**92.01** (+0.72)	**74.19** (+0.85)	81.88 -	**80.64** (+1.35)	**93.49** (+0.35)	**91.47** (+0.60)	**84.59** (+0.99)

CTCFNet [63]). To ensure consistency, all experiments in this section are performed in the same experimental environment and hyperparameter settings.

1) Results on the Vaihingen Dataset: Table VII demonstrates the quantitative results of all methods on the Vaihingen dataset. The middle section presents the Intersection-over-Union (IoU) values for each individual category, while the right-hand side of the table showcases the overall evaluation metrics. By comparing the metrics of OA, m$F1$, and mIoU, it can be clearly discerned that the LCCN outperforms other methods. PSPNet and DeepLabV3+ employ pooling and dilated convolutions to acquire more comprehensive context information and yield favorable outcomes. In comparison with BANet, MSGCNet attains a substantial enhancement in the IoU of each category. This implies that attention mechanism-based networks utilized for natural image segmentation might encounter difficulties in coping with the challenges of HRSIs. AANet integrates the channel context information with the spatial depth and leverages the category information to direct pixel classification, thereby attaining excellent performance on small objects. MSGCNet integrates a scale-aware module following the extraction of global context, thereby capturing more detailed information within the spatial domain. Consequently, it demonstrates a high recognition rate in the tree category. Moreover, LCCN attains the highest IoU in four semantic categories, attributed to the comprehensive capture of multiscale global context features.

To gain a more intuitive understanding of the segmentation performance of our proposed method, we present the visualization results of experiments conducted on the IRPRS Vaihingen dataset, as depicted in Fig. 7. Evidently, from Fig. 7, LCCN illustrates excellent performance on the Vaihingen dataset. By implementing multiscale hierarchical processing of feature maps, refining the information aggregation mechanism within the attention module, and adopting a comprehensive feature fusion approach, LCCN exhibits its proficiency in delivering robust performance even in intricate and complex semantic scenarios. Specifically, the regions circumscribed by ellipses in the initial two subfigures of Fig. 7 exhibit pronounced shadowing effects, which significantly impair the segmentation outcome. A multitude of existing methods erroneously segment these shadowed regions. In contrast, LCCN, endowed with superior semantic extraction capabilities, can generate more precise boundaries. This enables our method to achieve relatively high accuracy in segmenting the low vegetation and building categories. In the third sample image, the impervious surface bears a strong resemblance to a building, leading to misclassifications by other methods to some extent. By leveraging sufficient contextual information and achieving multiscale feature fusion, LCCN can accurately handle the similar features among different categories and generate precise boundaries.

2) Results on the Potsdam Dataset: In the IRPRS Potsdam dataset, the experimental results are shown in Table VIII. LCCN demonstrates the best results in terms of overall OA, m$F1$, and mIoU. Especially for the impervious surface and building categories, LCCN attains results of 88.61% and 93.36%. Among them, DC-Swin uses the Swin Transformer as the backbone and pays more attention to global information during the encoding stage. Through comparison, it can be

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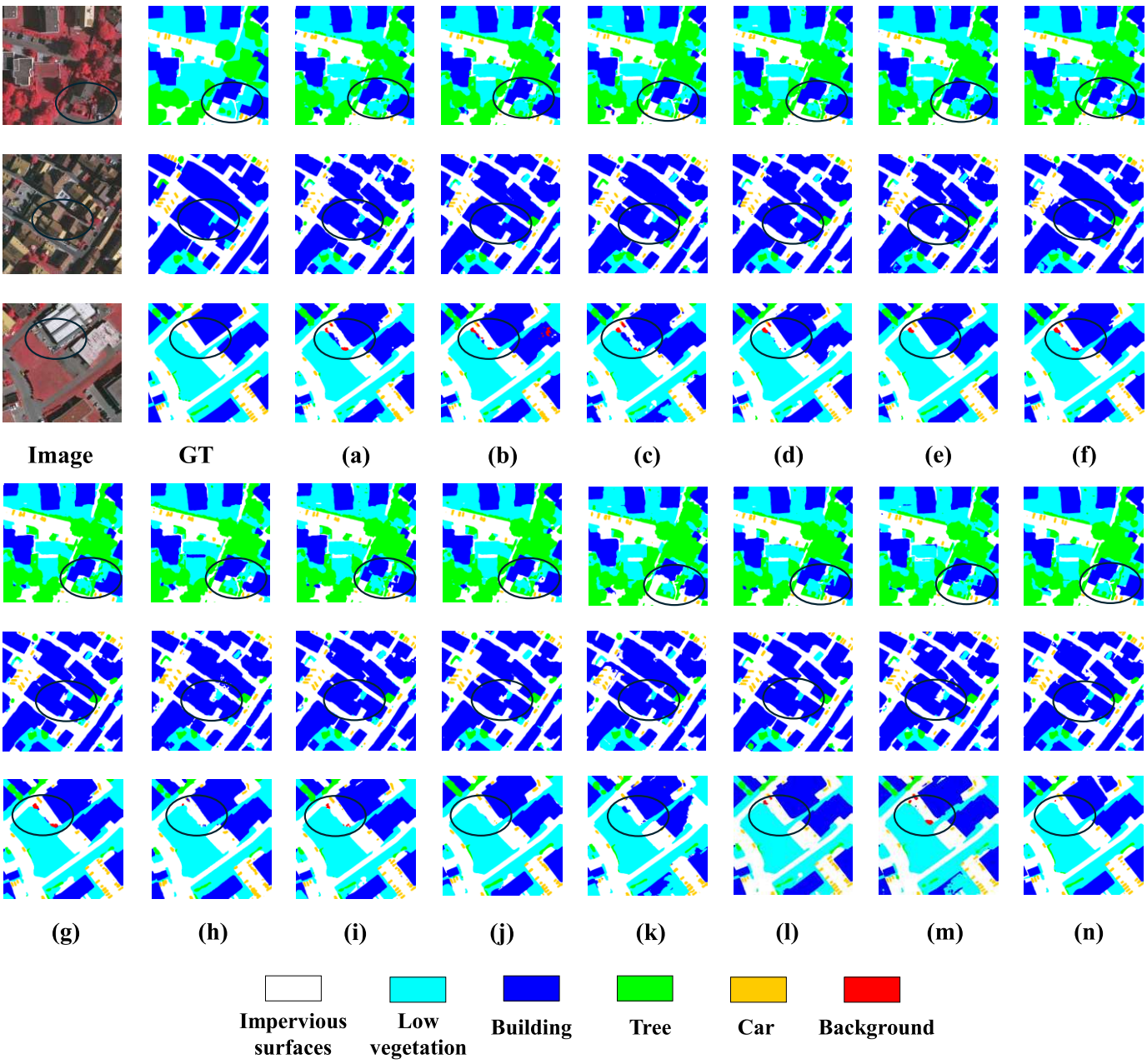


Fig. 7. Results of visualization experiments on the IRPRS Vaihingen dataset. (a)–(n) PSPNet, DeepLabV3+, MAResU-Net, MANet, A2-FPN, BANet, ABCNet, DC-Swin, UNetFormer, CMTFNet, DSNet, AANet, MSGCNet, and LCCN.

found that DC-Swin performs better than other methods in the impervious surface category with a larger area, reflecting the powerful global modeling ability of Swin Transformer. ABCNet adopts a bilateral architecture, introduces linear attention into a lightweight segmentation model to enhance the features output by the context path, and designs a feature aggregation module with linear attention as the core to fully aggregate the features output by the context and spatial paths, enhancing the network's perception ability. Therefore, ABCNet has good results in the low-vegetation category. Our method's mIoU for low vegetation is slightly lower than the best value, while, for the remaining categories, it achieves the best results. The excellent results indicate that our proposed method can effectively improve the performance of remote sensing semantic segmentation.

To more intuitively observe the segmentation performance of our method, we present the visual experimental results on the ISPRS Potsdam dataset, as shown in Fig. 8. As can be seen from Fig. 8, it is observed that LCCN achieves excellent performance on the Potsdam dataset. For example, in the regions circumscribed by ellipses in the first image, trees are closely connected to low vegetation. Other methods have misjudgments to a certain extent. The features obtained through the CDCM possess semantic information while retaining sufficient spatial details, making the segmentation of the right-hand side contour of the lawn in the first image

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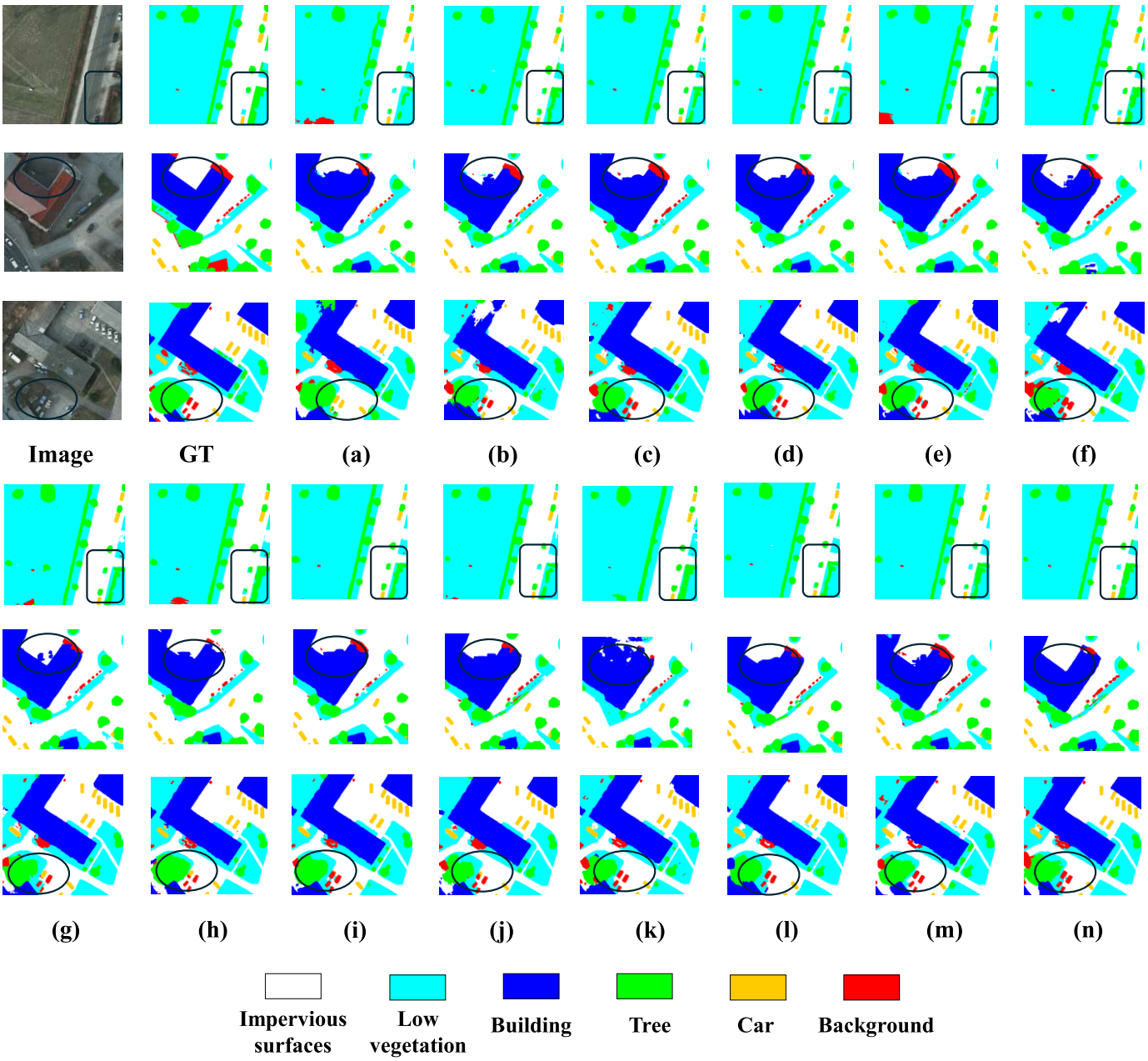


Fig. 8. Results of visualization experiments on the IRPRS Potsdam dataset. (a)–(n) PSPNet, DeepLabV3+, MAResU-Net, MANet, A2-FPN, BANet, ABCNet, DC-Swin, UNetFormer, CMTFNet, DSNet, AANet, MSGCNet, and LCCN.

clear. In the regions that are encircled by ellipses within the second image, many methods wrongly classify impervious surfaces as houses. The LCCN can accurately handle the similar features between classes and generate relatively accurate boundaries, achieving excellent results. In the third image, because of a certain similarity between the background and the car category, some methods misidentify the background category as cars. LMFAM fully integrates global and local features, which effectively alleviates the interference from complex backgrounds. Equipped with fine-grained context information, LCCN can accurately distinguish different categories.

3) Quantitative Comparison With Nonlightweight State-of-the-Art Networks: The experimental results in Table IX provide a comprehensive quantitative comparison between our proposed method and nonlightweight state-of-the-art networks on the IRPRS Vaihingen and Potsdam test sets. In terms of overall segmentation performance, our method achieves excellent results across core metrics. On the Vaihingen test set, the mIoU of our method reaches 84.59%, second only to FTUNetformer (84.67%) and outperforming most comparative methods. On the Potsdam test set, the mIoU of our method is 86.51%, which is comparable to the top-performing LoG-CAN (86.57%). This indicates that our method has achieved

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TABLE VIII

QUANTITATIVE COMPARISON RESULTS ON THE IRPRS POTSDAM TEST SET WITH STATE-OF-THE-ART LIGHTWEIGHT NETWORKS. THE BEST AND SECOND BEST VALUES IN THE COLUMN ARE IN BOLD AND UNDERLINED

Method	IoU(%)					OA(%)	F1(%)	mIoU(%)
	ImSurf	Building	LowVeg	Tree	Car			
PSPNet [11]	83.21	89.35	75.35	76.88	89.35	88.87	90.49	82.83
DeepLabV3+ [14]	85.00	90.43	75.98	77.51	89.74	89.42	91.03	83.73
MAResU-Net [56]	86.82	91.02	77.04	78.78	92.34	90.48	91.88	85.20
MANet [22]	85.70	91.84	77.06	79.27	91.77	90.32	91.85	85.13
A2-FPN [59]	87.89	92.63	77.13	79.19	91.55	90.97	92.16	85.68
BANet [57]	87.89	91.86	76.41	78.84	90.71	90.75	91.85	85.14
ABCNet [43]	86.23	92.34	**78.22**	79.84	92.10	90.77	92.21	85.75
DC-swin [37]	88.60	93.11	77.39	79.39	90.80	91.24	92.26	85.86
UNetFormer [44]	87.69	92.05	77.04	79.32	91.47	90.75	92.07	85.51
CMTFNet [46]	85.89	91.89	78.17	79.53	92.36	90.65	92.11	85.57
DSNet [35]	85.09	92.02	77.51	78.97	92.27	90.33	91.87	85.17
AANet [45]	87.67	91.93	77.19	79.73	91.80	90.84	92.16	85.67
MSGCNet [58]	87.63	92.74	77.21	79.34	91.48	90.87	92.16	85.68
Ours	**88.61** (+0.01)	**93.36** (+0.25)	78.10 -	**80.03** (+0.19)	**92.46** (+0.10)	**91.50** (+0.26)	**92.64** (+0.38)	**86.51** (+0.65)

TABLE IX

QUANTITATIVE COMPARISON RESULTS ON THE IRPRS VAIHINGEN (VALUES ON THE LEFT) AND POTSDAM (VALUES ON THE RIGHT) TEST SET WITH NON LIGHTWEIGHT STATE-OF-THE-ART NETWORKS. THE BEST AND SECOND BEST VALUES IN THE COLUMN ARE IN BOLD AND UNDERLINED

Method	Backbone	IoU(%)					OA(%)	F1 (%)	mIoU(%)	Parameters (M)	FLOPs (G)
		ImSurf	Building	LowVeg	Tree	Car					
WiCoNet[60]	ResNet-50	86.63/85.83	91.03/92.58	72.42/77.20	81.65/78.57	80.03/90.62	90.75/90.23	90.19/91.75	82.35/84.96	38.24	125.94
DC-swin[37]	Swin-Small	93.40/88.60	91.05/93.11	71.48/77.39	81.53/79.39	78.42/90.80	92.88/91.24	90.69/92.26	83.18/85.86	66.90	267.79
EMRT[61]	ResNet-50	87.23/86.18	91.52/92.55	72.38/77.21	81.28/79.76	80.27/91.04	90.91/90.51	90.29/91.98	82.54/85.35	54.85	142.89
LoG-CAN[62]	ResNet-50	87.66/**88.81**	91.90/93.86	74.12/**78.54**	**82.78**/**80.20**	81.60/92.44	91.45/91.37	90.96/**92.68**	83.61/**86.57**	30.91	49.51
CTCFNet[63]	PVT+CNN	88.37/86.92	**92.10**/93.62	74.26/76.94	82.76/79.46	81.56/91.92	91.64/90.68	91.07/92.20	83.81/85.77	74.71	188.21
FTUNetformer[44]	Swin-Base	94.16/87.74	92.07/**93.92**	73.38/77.91	82.08/79.53	**81.65**/**92.87**	93.46/91.09	**91.51**/92.56	**84.67**/86.39	96.14	383.90
Ours	ResNet-18	**94.23**/88.61	92.01/93.36	74.19/78.10	81.88/80.03	80.64/92.46	**93.49**/**91.50**	91.47/92.64	84.59/86.51	**11.57**	**47.96**

an advanced level of accuracy in remote sensing image segmentation tasks. In addition, the OA (93.49%/91.50%) and $F1$ score (91.47%/92.64%) of our method also remain at a high level, further verifying the robustness of our method.

Our method has significant advantages in key categories. In the ImSurf category, it achieves the highest IoU (94.23%) on the Vaihingen test set. It also performs excellently in the segmentation of buildings (92.01%/93.36%) and cars (80.64%/92.46%), among which the segmentation performance of the car category on the Potsdam test set is comparable to that of LoG-CAN. These results show that our model can not only effectively capture large-area impervious surfaces but also accurately identify small targets such as vehicles. For the Tree and LowVeg categories, the IoU of our method (81.88%/80.03% and 74.19%/78.10%, respectively) is slightly lower than that of leading methods, suggesting that there is room for improvement in distinguishing complex vegetation textures.

Notably, our method only requires 1/3 to 1/8 of the parameters and FLOPs of methods such as DC-Swin, CTCFNet, and FTUNetformer. This significant difference highlights that our model achieves leading accuracy while maintaining lightweight characteristics, resolving the tradeoff between accuracy and computational cost. The experimental results confirm that our method, through optimized lightweight design and effective feature modeling, achieves an optimal balance between segmentation accuracy and efficiency and is superior to or comparable in performance to more complex state-of-the-art networks on both test sets. This fully demonstrates its practical value in remote sensing image analysis.

4) Feature Visualization: We adopted the Grad-CAM-based method [64] to demonstrate the impact of different settings of

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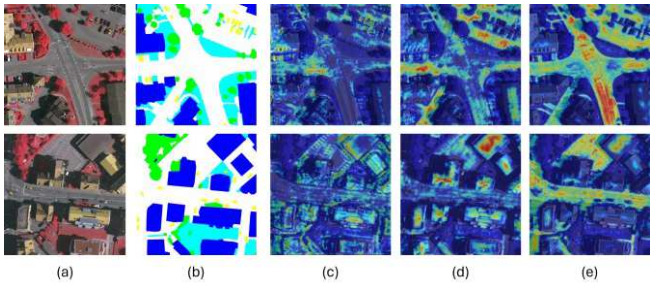


Fig. 9. Attention maps. (a) Images. (b) Ground truth. (c) Baseline. (d) Baseline + CDCM. (e) Baseline + CDCM + LFMAM.

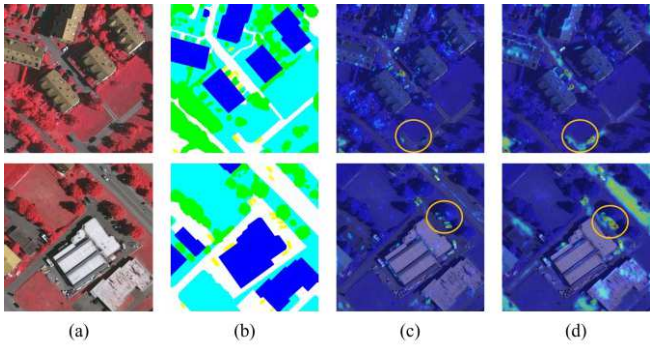


Fig. 10. Visualizations of the output features of the "car" and "clutter" classes by the baseline and baseline + LFMAM. (a) Images. (b) Ground truth. (c) Baseline. (d) Baseline + LFMAM.

the CDCM module and the LFMAM module on the attention heatmaps. Fig. 9 displays the attention maps generated by the proposed model on the Vaihingen dataset as each component of the method is incrementally incorporated. It can be observed that the baseline (ResNet-18) pays high attention to local details but less attention to global features [see Fig. 9(c)]. After incorporating the CDCM module [see Fig. 9(d)], the model's attention to multiscale feature targets (such as buildings and low vegetation) is significantly enhanced. Subsequently, the LFMAM is introduced, which effectively integrates the extraction of local and global information. By comparing the attention maps in Fig. 9(e) with those in Fig. 9(c) and (d), it is evident that our proposed method can better capture contextual information and local details.

As shown in Fig. 10, the comparison of attention heatmaps for shadowed areas (such as dark regions formed by tree occlusions) and small targets (cars). The results show that through global–local feature fusion, LFMAM enables the attention weights of small target regions to be significantly higher than those of background regions, which further verifies the adaptability of the module in complex scenarios.

VI. Conclusion

This article introduces a novel LCCN based on an innovative encoder–coupler–decoder architecture, aiming to address the challenges of significant object scale variations and susceptibility to complex background interference in remote sensing images. CDCM employs feature grouping and loop aggregation to perform multiscale feature extraction and global information modeling on each subfeature map. By selectively focusing on feature channels and spatial information across diverse dimensions, it reduces computational overhead while extracting richer spatial feature information. In addition, LFMAM proposes a more comprehensive feature mapping structure, which promotes the synergistic effect among the features of different branches, further refines the feature correlations and differences, and significantly reduces the computational complexity at the same time, making it more lightweight. Compared with multiple state-of-the-art lightweight segmentation models, LCCN demonstrates superior segmentation accuracy and achieves a favorable balance between computational cost and precision. This innovative architecture not only showcases its robustness in HRSI processing but also provides new avenues for future research. In subsequent studies, we endeavor to explore additional feature extraction and context modeling methods to further optimize the model structure and advance the performance of semantic segmentation in remote sensing images.

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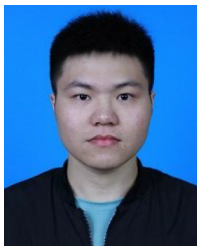
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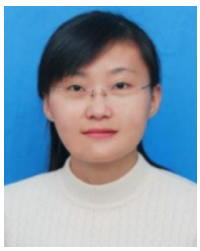
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